

Effect of a high nutrient density diet on long-term weight loss: a retrospective chart review.

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BACKGROUND: A high nutrient density (HND) vegetable-based diet offers a dietary model extremely low in saturated fat as well as refined carbohydrates and emphasizes a liberal intake of fresh fruits, vegetables, beans, and nuts. We conducted a retrospective chart review of patients who came to a family practice office seeking nutritional counseling for weight loss. All of these patients were prescribed an HND diet in an extended counseling session with a family physician. **METHODS:** A convenience sample (N = 56) of all patients seeking dietary counseling for weight loss from a family practice physician in a 3-year period was included in the chart review. No personal identifying data were recorded. The initial counseling sessions averaged 1 hour in length. Patients were provided with a sample HND daily meal plan and recipes and with verbal and written information about the rationale for the diet. Data recorded from patients' charts at 6-month intervals for up to 2 years of follow-up (when available) included weight, blood pressure, total cholesterol, high-density lipoprotein (HDL) cholesterol, low-density lipoprotein (LDL) cholesterol, triglycerides, and cholesterol:HDL ratio. Non-parametric statistical testing using the Friedman rank order (exact) test for k-related samples was conducted. A follow-up survey on adherence and medication use was completed by 38 patients. **RESULTS:** Of the 33 patients who returned for follow-up after 1 year, the mean weight loss was 31 lbs (P = .000). Of the 19 patients who returned after 2 years, the mean weight loss was 53 lbs (P = .000), mean cholesterol fell by 13 points, LDL by 15 points, triglycerides by 17 points, and cardiac risk ratio dropped from 4.5 to 3.8. Changes in systolic and diastolic blood pressure were highly significant at all follow-up time intervals (P < or = .001). There was a significant correlation between adherence and degree of weight loss (P = .011). **CONCLUSIONS:** Weight loss was sustained in patients who returned for follow-up and was more substantial in those who reported good adherence to the recommendations. However, many patients were lost to follow-up. Favorable changes in lipid profile and blood pressure were noted. An HND diet has the potential to provide sustainable, significant, long-term weight loss and may provide substantial lowering of cardiac risk in patients who are motivated and provided with extended one-on-one counseling and follow-up visits. Development of tools to aid in patient retention is an area for possible further study. Clinical trials with long-term follow-up are needed to further test the therapeutic potential and to examine adherence and follow-up issues related to this dietary approach. An HND diet as demonstrated with this group may be the most health-favorable and effective way to lose weight for appropriately motivated patients.